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TECHNICAL MEMORANDUM

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COMPARISON OF THE EFFECTIVENESS OF FLARES WITH THAT OF FINS FOR STABILIZING LOW-FINENESS-RATIO BODIES AT MACH NUMBERS FROM 0.6 TO 5.8

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May 1962

INTRODUCTION

The relative effectiveness of flares as compared with fins for stabilizing bodies of low fineness ratio is of interest in the design of ballistic and atmospheric entry vehicles. Although many investigations have been made of the effectiveness of flares for stabilizing bodies (e.g., refs. 1 through 10), there is a need for direct comparisons of flare and fin effectiveness over a wide Mach number range. Accordingly, an investigation was conducted to determine the stability and drag of several flared and finned bodies at Mach numbers from 0.6 to 5.8. Models consisting of conical noses with cylindrical midsections and finned or flared aftersections were tested. This study included investigation of the effects of flare angle (from 0° to 20°) and of nose and fin bluntness. The range of flare angles chosen enabled normal-force and stability comparisons of finned and flared models to be made on the basis of either specified equal plan-form area or equal drag. Both experimental and analytical comparisons have been made, and the purpose of this report is to discuss these comparisons.

A 5 7 5

NOTATION

A cross-sectional area of cylindrical portion of body

a fin semispan measured from surface of cylindrical portion of body

C_D drag coefficient (exclusive of base drag) at zero angle of attack

C_{m_α} pitching-moment-curve slope referred to the moment centers shown in figures 8, 9, and 12,

$$\frac{d}{d\alpha}\left(\frac{\text{pitching moment}}{q_\infty A d}\right)$$

C_{N_α} normal-force-curve slope, $\frac{d}{d\alpha}\left(\frac{\text{normal force}}{q_\infty A}\right)$

$C_{p_{stag}}$ stagnation pressure coefficient, $\frac{p_{stag} - p_\infty}{q_\infty}$

d diameter of cylindrical portion of body

d' body base diameter

d_w	trip wire diameter
E	complete elliptic integral of second kind
K	ratio of normal-force component to normal force of fin alone
l	length
M	free-stream Mach number
p_{stag}	stagnation pressure
p_∞	free-stream static pressure
q_∞	free-stream dynamic pressure
R	Reynolds number based on d
r	radius of fin leading edge
r_S	radius of spherical portion of nose
x_{cp}	center of pressure measured rearward from nose-cylinder juncture
x'_{cp}	center of pressure measured rearward from moment centers shown in figures 8, 9, and 12
α	angle of attack, deg
β	$\sqrt{M^2 - 1}$
δ	half-angle, deg

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Subscripts

A	aftersection
B(W)	body in presence of fins
C	unblunted cone
cy	cylinder
F	fins
Fl	flare

N	nose
S	spherical portion of nose
W(B)	fins in presence of body

EXPERIMENTAL CONSIDERATIONS

Wind Tunnels

The experimental investigation was conducted in the Ames 2- by 2-Foot Transonic Wind Tunnel and the 1- by 3-Foot Supersonic Wind Tunnel. The 2- by 2-foot tunnel is of the closed-circuit, variable-pressure type. It has a flexible-plate nozzle and a perforated test section which permits continuous choke-free operation at Mach numbers up to 1.4. The 1- by 3-foot tunnel is also a closed-circuit, variable-pressure type and has a flexible-plate nozzle that provides a variation of Mach number from about 1.4 to 6. In these tunnels the Reynolds number is changed by varying the total pressure within the approximate limits of 1/5 of an atmosphere to 4 atmospheres.

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The water content of the air in the tunnels is maintained at less than 0.0003 pound of water per pound of dry air. Consequently, any effect of humidity on the flow is negligible.

Models

In order to determine the stabilizing effectiveness of flares and fins, models consisting of bodies with and without flared and finned aftersections were tested. Sketches and designations of the models are shown in figure 1 in three groupings: nose shapes and bodies (fig. 1(a)), bodies with flares (fig. 1(b)), and bodies with cruciform fins (fig. 1(c)). In addition, the sizes and positions of wire rings that were placed on the body noses to trip the boundary layer are shown in figure 1(d). Two nose shapes were used: one, a sharp cone of $14^{\circ}2'$ semiapex angle (designated N_S), and the other, a spherically blunted cone ($2r_S/d = 0.50$) with the same cone angle (designated N_B). These nose pieces were constructed for testing both separately and with cylindrical bodies of 2 and 3.37 diameters in length (designated C and C_L, respectively). The cylindrical bodies were constructed so that flared or finned aftersections could be attached. All of the flared aftersections were of equal length (1.373 diameters) but with flare angles of 0°, 10°, and 20° (fig. 1(b)). The finned aftersections were of the same length and plan form as the 20° flared aftersection. As noted in figure 1(c), cruciform fins with all sharp or all blunt leading edges and combinations of sharp and blunt

leading edges were employed. In addition, a model with blunt fins superimposed on a 10° flare was tested. All models were sting mounted from the rear with a strain-gage balance positioned inside each model. Photographs of typical flared and finned models mounted in the 1- by 3-foot supersonic wind tunnel are shown in figure 2.

Tests

A 5 7 5 Tests were made to determine the normal-force-curve slope, center of pressure, drag, and boundary-layer-flow characteristics of the models at zero angle of attack for various Mach numbers. The test Mach numbers and Reynolds numbers based on body diameter are summarized in the following table:

M	$R \times 10^{-6}$	Tunnel
0.60 to 1.40	0.38	2- by 2-foot
1.37 to 4.06	.49	1- by 3-foot
5.11	.35	1- by 3-foot
5.82	.18	1- by 3-foot

To determine accurately the normal-force-curve slopes and centers of pressure at zero angle of attack, the models were tested at half degree increments throughout the α range from -4° to +4°. The blunt-nosed bodies alone and in combination with the flared and finned aftersections were tested throughout the entire Mach number range (M = 0.60 to 5.82). The sharp-nosed models and the models with combined sharp and blunt fins ($N_BCF_{B_VS_H}$ and $N_BCF_{B_HS_V}$) were tested only at selected supersonic Mach numbers. All of the models were tested with wire rings mounted on their noses in an effort to trip the boundary-layer flow (fig. 1(d)). The blunt-nosed model with the 20° flare (N_BCFl_{20}) was also tested without any rings at Mach numbers from 2.94 to 5.81. For all tests the shadowgraph technique was employed to observe model boundary-layer flow.

Reduction and Precision of Data

All of the force and moment data have been reduced to coefficient form. From plots of normal-force coefficient and pitching-moment coefficient versus angle of attack, values of C_{N_α} and C_{m_α} were determined, and center-of-pressure positions (x_{cp}/d) were computed. The measured base pressures were used to compute the base drag which was subtracted from the total drag measurements, so that the drag coefficients presented are for forces ahead of the body base.

The precision of the final data is affected by uncertainties in the measurement of the forces and moments, and in the determination of the stream static and dynamic pressures used in reducing the forces and moments to coefficient form. These individual uncertainties led to the following estimated uncertainties:

C_{N_α}	±0.0015 per deg	α	±0.1°
x_{cp}/d	±0.04	M	±0.01 for $M \leq 2.94$
C_D	±0.015	M	±0.03 for $M > 2.94$

RESULTS AND DISCUSSION

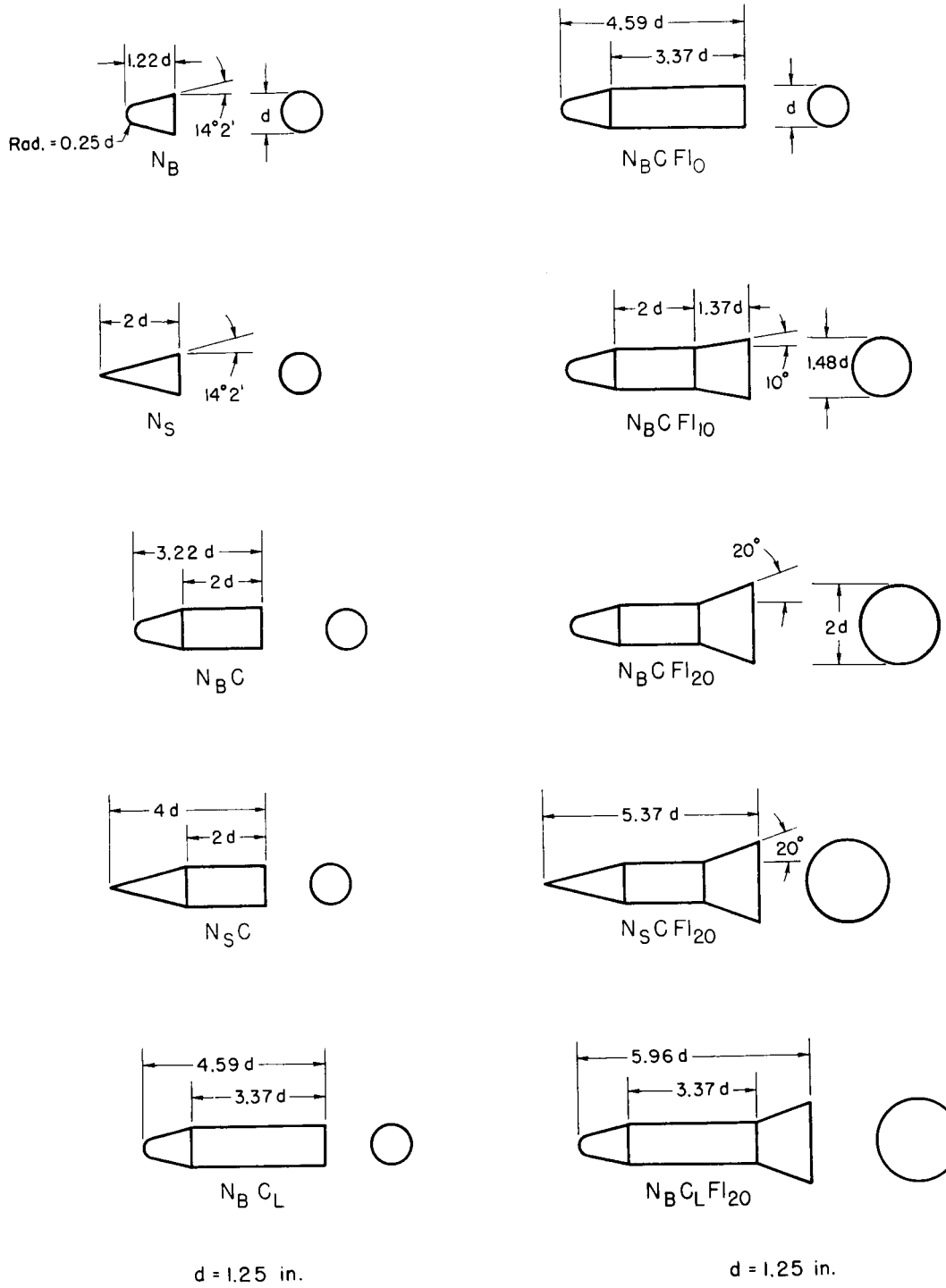
For all the models tested, experimental values of C_{N_α}, x_{cp}/d, and C_D as functions of Mach number are presented in figures 3 through 6. Results for the nose shapes and bodies are given in figure 3, while those for the bodies with flares and for the bodies with fins are given in figures 4 and 5, respectively. In figure 6 results are compared for the flared and finned bodies of equal plan-form area. Data for the models having boundary-layer trip rings are designated by unflagged symbols, whereas results for the models without trip rings are designated by flagged symbols. Representative shadowgraphs are shown in figure 7. Reference is made to these figures in the following discussion which is divided into two major sections: (1) comparison of experimental effectiveness of flares and fins and (2) comparison of computed with experimental results.

Comparison of Experimental Effectiveness of Flares and Fins

Flared and finned models of equal plan-form area.- Throughout the Mach number range, the results clearly show that flares are more effective than fins of equal plan-form area. For example, in figure 6 it is seen that the values of C_{N_α} (fig. 6(a)) for the flared models were higher at supersonic speeds than those for comparable finned models, and the centers of pressure (fig. 6(b)) were more rearward. As a result, the flare provided considerably more stability than the fins, especially at the higher Mach numbers. The gains in effectiveness of the flared over the finned models were, of course, accompanied by much greater drag for the flared models (fig. 6(c)).

It is important to note here that some of the aerodynamic characteristics for the 20° flared models were greatly influenced by flow separation ahead of the flare. The occurrence of flow separation was

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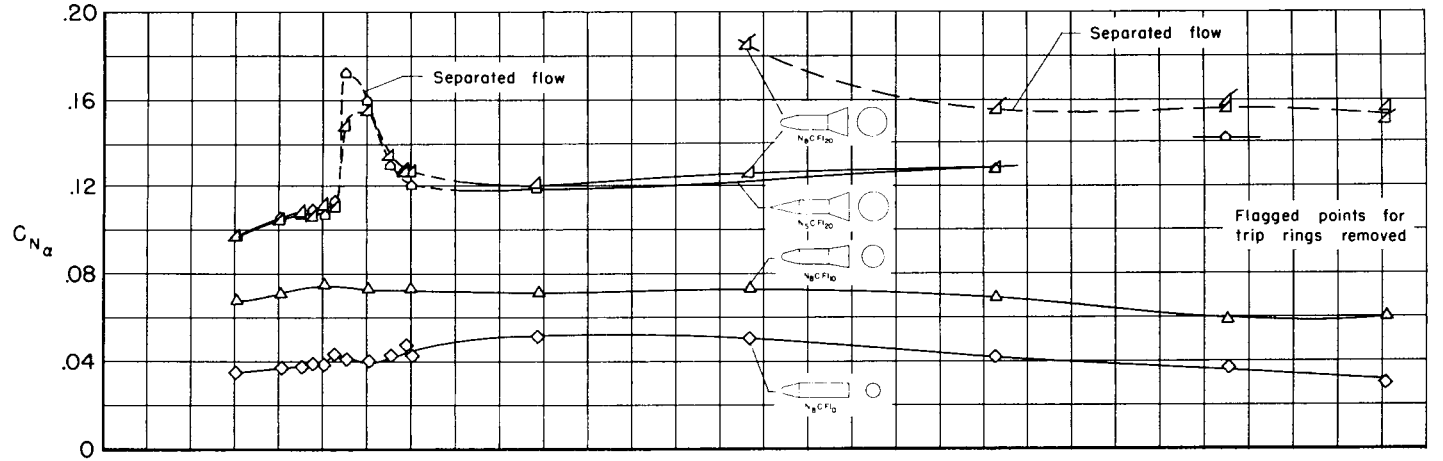


(a) Nose shapes and bodies.

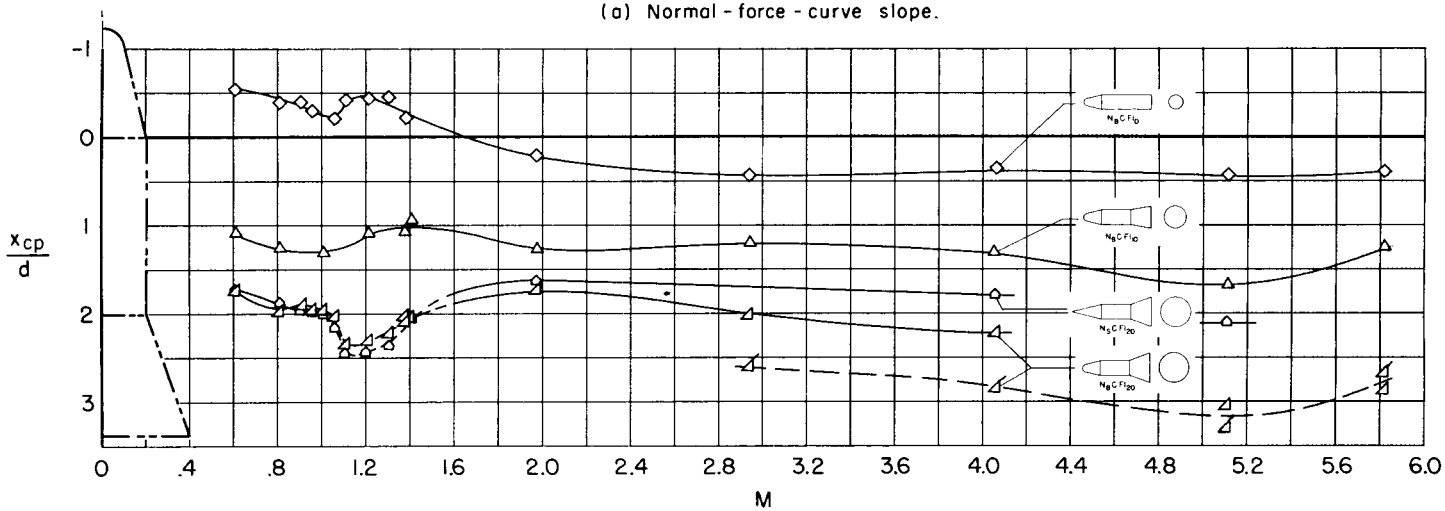
(b) Bodies with flares.

Figure 1.- Models tested.

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(a) Normal-force-curve slope.



(b) Center of pressure.

Figure 4.- Effect of flare angle on aerodynamic characteristics of flared bodies; $\alpha = 0^\circ$.